

GARCH, explained on BTC

Why returns aren't iid, what GARCH does about it, and what it buys you for a 5-minute market

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1. The problem GARCH solves

Take BTCUSDT (Binance perp) mid-price log returns on a 1-minute grid. Two facts, both visible in Figure 1:

1. **The sign of the next return is unpredictable.** The autocorrelation of r_t is zero at every lag (Figure 2, left). Returns are a martingale difference: uncorrelated, mean zero.
2. **The size of the next return is predictable.** Big moves cluster. The autocorrelation of r_t^2 starts at 0.2 and is still positive two hours later (Figure 2, right).

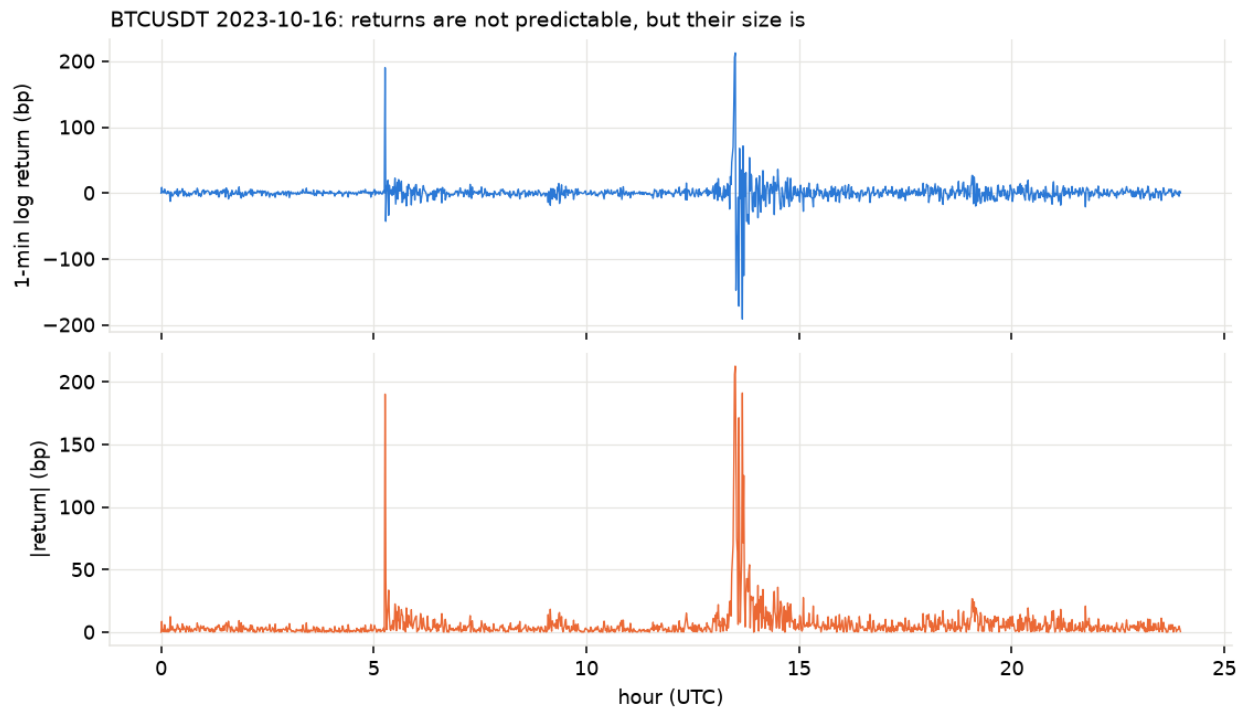


Figure 1: A day of 1-minute returns. The top panel looks like noise; the bottom panel shows the noise has a volume knob that somebody keeps turning.

averaged over 125 days; grey band = 95% noise level for one day

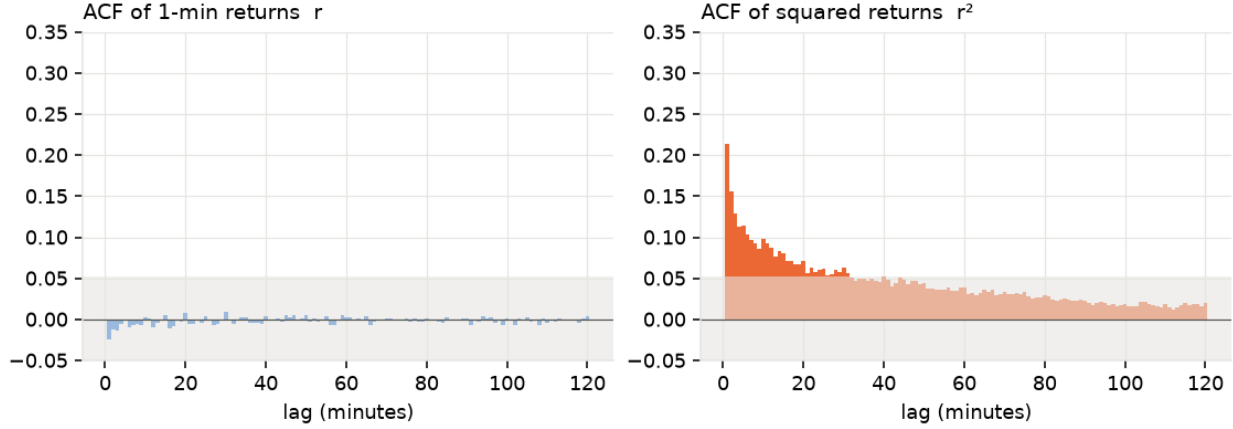


Figure 2: Left: returns are uncorrelated. Right: squared returns are strongly and persistently autocorrelated. This pair of pictures is the entire motivation for GARCH.

Uncorrelated but not independent. Any model with iid returns (a random walk with constant σ , for instance) gets the first fact right and the second one completely wrong, and the consequence shows up in the tails: summed over k steps, iid returns have excess kurtosis κ_1/k , which for BTC would predict ≈ 0.7 at 5 minutes. The measured value over 495 days is ≈ 120 .

The “volume knob” picture

The way to hold this in your head is

$$r_t = \sigma_t \varepsilon_t$$

where ε_t is a standardised shock (mean 0, variance 1, roughly iid) and σ_t is the *current* volatility - a slowly moving, persistent quantity. The shock is the coin flip; σ_t is how much money is riding on it. A 5-minute return is $\sum \sigma_s \varepsilon_s \approx \sigma_t \sum \varepsilon_s$, i.e. one Gaussian-ish draw scaled by whatever σ happened to be in force. Pool a year of those and you get a **mixture of Gaussians with different widths** - a sharp peak (calm windows) with fat tails (wild windows). The CLT never gets to work because within any short window you are stuck inside one σ regime.

GARCH is nothing more than a specific, fittable recipe for σ_t .

2. The model

ARCH (Engle, 1982)

The simplest recipe: today’s variance is a weighted average of recent squared returns,

$$\sigma_t^2 = \omega + \alpha_1 r_{t-1}^2 + \alpha_2 r_{t-2}^2 + \cdots + \alpha_q r_{t-q}^2.$$

If yesterday’s move was big, today’s expected move is big. It works, but you need many lags to capture the slow decay in Figure 2, and every lag is a parameter.

GARCH(1,1) (Bollerslev, 1986)

Add one lag of the variance itself:

$$\sigma_t^2 = \omega + \alpha r_{t-1}^2 + \beta \sigma_{t-1}^2$$

Three parameters. Read it as: *new variance = a constant + a bit of the latest surprise + most of the old variance*. Substituting recursively,

$$\sigma_t^2 = \frac{\omega}{1-\beta} + \alpha \sum_{j=1}^{\infty} \beta^{j-1} r_{t-j}^2,$$

so GARCH(1,1) is an **exponentially-weighted moving average of squared returns**, with decay β per step, plus a constant that pulls it back toward a long-run level. It is ARCH with infinitely many lags and geometrically decaying weights, for the price of one extra parameter.

Useful quantities that follow directly:

quantity	formula	meaning
long-run variance	$\bar{\sigma}^2 = \omega / (1 - \alpha - \beta)$	where σ_t^2 mean-reverts to
persistence	$\alpha + \beta$	fraction of a variance shock that survives one step
half-life	$\ln 0.5 / \ln(\alpha + \beta)$	steps for a shock to decay by half
reactivity	α	how much one surprise moves the estimate
h -step forecast	$\bar{\sigma}^2 + (\alpha + \beta)^{h-1}(\sigma_{t+1}^2 - \bar{\sigma}^2)$	geometric decay to the long-run level
kurtosis (Gaussian ε)	$3 \frac{1 - (\alpha + \beta)^2}{1 - (\alpha + \beta)^2 - 2\alpha^2}$	fat tails from vol variation alone

Constraints: $\omega > 0$, $\alpha, \beta \geq 0$ for positivity; $\alpha + \beta < 1$ for a finite long-run variance. When $\alpha + \beta = 1$ the model is IGARCH: variance shocks never decay and the long-run level is undefined (the RiskMetrics EWMA is IGARCH with $\omega = 0$, $\beta = 0.94$).

What it looks like

Figure 3 simulates 5 000 steps of iid Gaussian noise and of GARCH(1,1) with the *same* unconditional variance. The GARCH path has visible bursts and lulls; its histogram is peaked with heavy tails (excess kurtosis 3.3 from the formula above) even though every individual ε_t is Gaussian. That is fat tails manufactured purely by vol clustering.

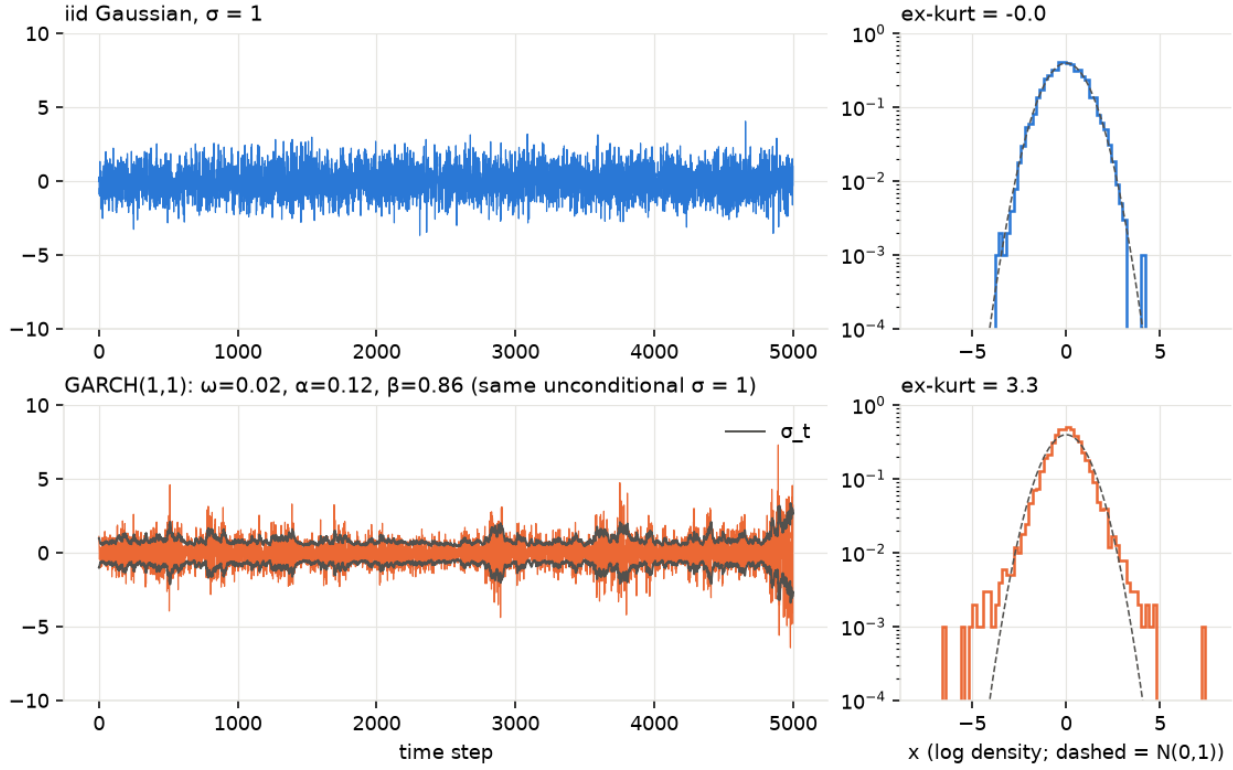


Figure 3: Same long-run variance, very different behaviour. The grey line in the lower left panel is $\pm\sigma_t$: the model's own volume knob.

3. Fitting it

The parameters come from maximum likelihood. Given (ω, α, β) , run the recursion forward to get every σ_t^2 , then the (Gaussian) log-likelihood is

$$\ell = -\frac{1}{2} \sum_t \left[\ln(2\pi\sigma_t^2) + \frac{r_t^2}{\sigma_t^2} \right],$$

and you maximise it numerically. The Gaussian likelihood gives consistent estimates of (ω, α, β) even when the true shocks are fat-tailed (this is “QMLE”); the shape of ε can be fitted afterwards to the standardised residuals $z_t = r_t/\sigma_t$.

On BTC 1-minute returns

Fitted on 291 consecutive days (June 2023 - March 2024, 419k minutes) of 1-minute mid returns:

	value
α	0.082
β	0.916
$\alpha + \beta$	0.9976
half-life	283 min (≈ 4.7 h)
long-run σ	6.6 bp / min ($\approx 40\%$ annualised)
Student-t ν for z	4.9

A note on how these were obtained, because it matters in practice. Unconstrained MLE at 1-minute frequency pushes $\alpha + \beta$ to 1 (IGARCH): the likelihood keeps improving as persistence goes to 0.9999, at which point the “long-run variance” comes out at 5x the sample variance and is meaningless. This is a well-known artefact of high-frequency data: the vol regime drifts over weeks and months, and the only way a GARCH(1,1) can track a drifting level is to give up on mean reversion. The fix used here is **variance targeting** - pin ω so the long-run variance equals the sample variance and estimate only α, β - which costs almost nothing in likelihood and gives an interpretable model. (Fitting on one contiguous era rather than days sampled across three years matters for the same reason.)

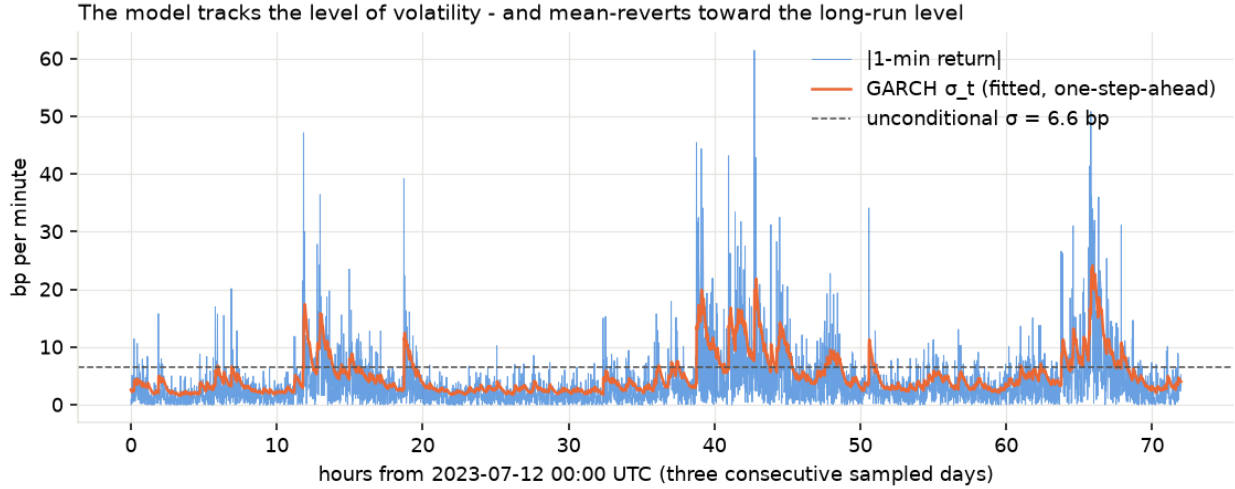


Figure 4: Fitted one-step-ahead σ_t against realised $|r_t|$ over three days. The estimate rises within a minute or two of a burst and takes hours to come back down.

Does it work?

The job of the model is to make $z_t = r_t/\sigma_t$ look iid. Three checks, before and after:

	raw r_t	$z_t = r_t/\sigma_t$	iid Gaussian
lag-1 autocorrelation of squares	0.37	0.003	0
excess kurtosis	240	101	0
$P(z > 3)$	1.5%	1.2%	0.27%
$P(z > 5)$	0.34%	0.17%	0.00006%

The clustering is gone: the squared standardised residuals are white. The tails are only half gone. A 5σ minute still happens once every ~ 600 minutes instead of once every 1.7 million. Figure 5 shows why: the standardised residuals are not Gaussian at all - they sit on a Student-t with about 5 degrees of freedom. This is the normal state of affairs. GARCH with a Gaussian ε explains kurtosis of order 3-10; the rest has to come from a fat-tailed ε (Student-t is the default) or an explicit jump component.

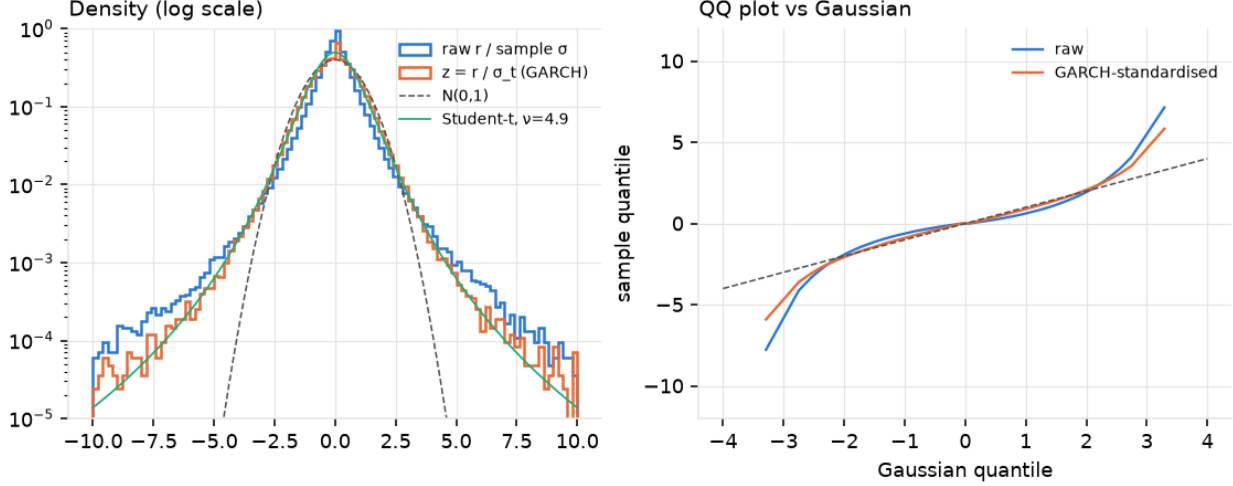


Figure 5: Standardising by the GARCH σ_t pulls the distribution in (orange vs blue) but leaves it far from Gaussian (dashed). A Student-t with $\nu \approx 5$ (green) fits the residual shape well.

4. Forecasting

The one-step forecast is just the recursion: $\hat{\sigma}_{t+1}^2 = \omega + \alpha r_t^2 + \beta \sigma_t^2$. Multi-step forecasts decay geometrically toward the long-run level,

$$E_t[\sigma_{t+h}^2] = \bar{\sigma}^2 + (\alpha + \beta)^{h-1} (\hat{\sigma}_{t+1}^2 - \bar{\sigma}^2),$$

and the forecast variance of a k -step return is the sum of those. Figure 6 shows the term structure from two starting states. Two things to notice:

- With a half-life of ~ 5 hours, **the 5-minute-ahead vol is essentially the current vol**. For a 5-minute market the mean-reversion term is irrelevant; what matters is getting σ_t right *now*.
- After a burst, the forecast for the next few hours stays elevated. The model's memory is what makes it useful; it is also what makes it slow to notice the burst is over.

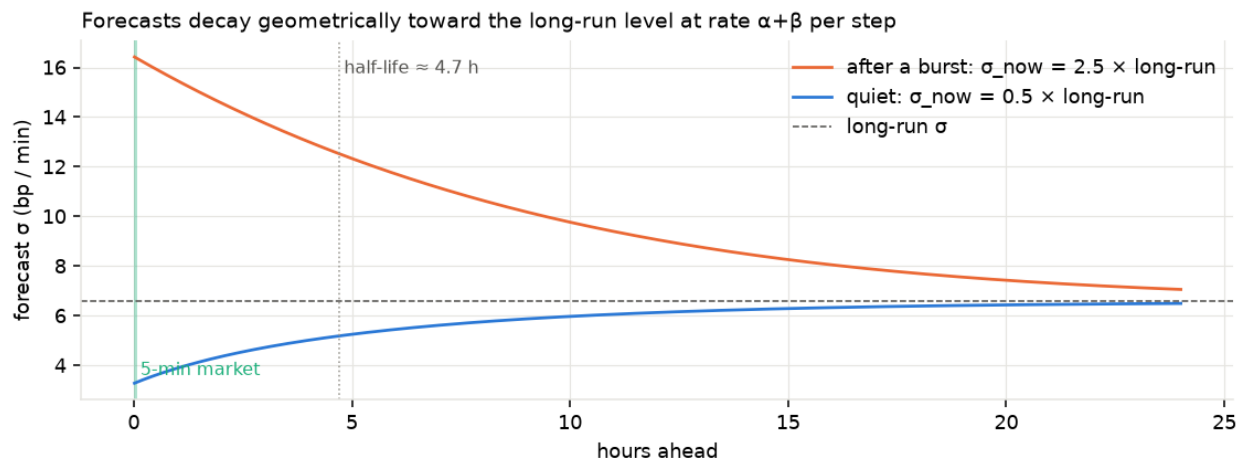


Figure 6: Variance forecasts from a high-vol and a low-vol starting point. The green sliver at the left is the 5-minute horizon: for that purpose the forecast is flat.

5. What plain GARCH gets wrong, and the usual fixes

Intraday seasonality. BTC trades 24/7 but not evenly: 1-minute vol at 13:00-16:00 UTC is double the 04:00-10:00 level (Figure 7). GARCH treats that predictable rise as a “shock” every afternoon and then slowly forgets it every night. Standard fix: estimate the time-of-day profile $s(\tau)$, fit GARCH to $r_t/s(\tau_t)$, multiply back for forecasts.

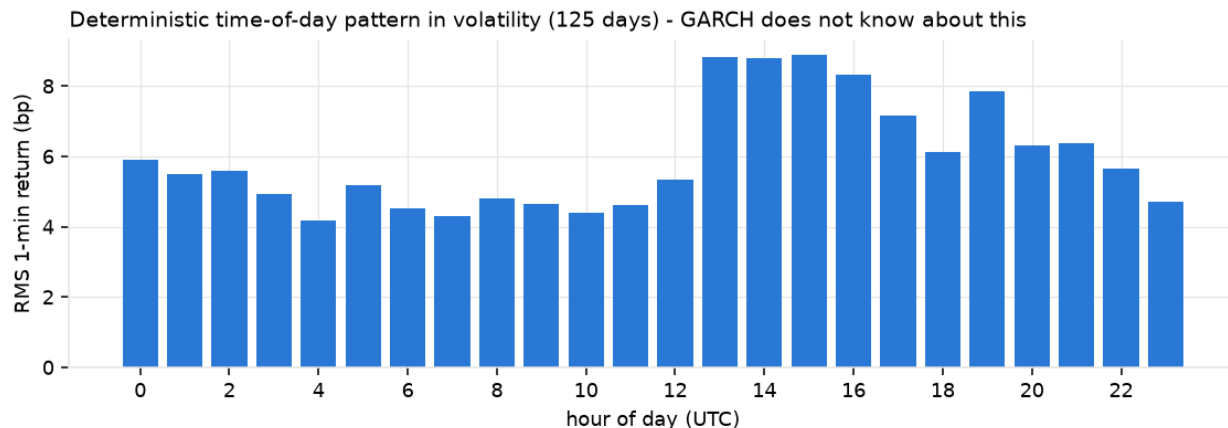


Figure 7: Root-mean-square 1-minute return by hour of day, over 125 days. This is deterministic and known in advance, and a plain GARCH ignores it.

Leverage / asymmetry. In equities negative returns raise vol more than positive ones; GJR-GARCH adds a term $\gamma r_{t-1}^2 \mathbf{1}[r_{t-1} < 0]$, EGARCH models $\ln \sigma^2$ with a sign term. For BTC at intraday horizons the asymmetry is weaker and both directions of liquidation cascade exist, but it is worth testing.

Reaction speed. GARCH updates once per bar using one squared return - a noisy signal. If you have finer data, a *realised variance* computed from many sub-bar returns (e.g. the sum of squared 1-second returns over the last 15 minutes) is a far better vol estimate than anything a 1-minute GARCH can extract from 1-minute bars. On this dataset, scaling 5-minute returns by trailing 15-minute realised vol from 1-second data cut excess kurtosis from 170 to ~50, better than the 1-minute GARCH manages.

Models built around this idea are HAR-RV (Corsi, 2009) and realised-GARCH (Hansen et al., 2012). GARCH is what you use when you *only* have the bars.

Jumps. No vol model removes the 0.2% of windows that are news or cascades. Those need a fat-tailed innovation ($\nu \approx 3-5$) or an explicit jump process, and the tails of the prediction should be read as “at least this fat”.

6. What this means for a 5-minute BTC binary

The probability that BTC ends the window above the strike is, under any of these models, essentially a function of

$$d = \frac{\ln(K/P_t)}{\hat{\sigma} \sqrt{T-t}},$$

the distance to the strike in forecast-vol units. Everything above is about producing $\hat{\sigma}$ and the shape of the distribution around it:

1. **Level.** Use a fast, floored estimate of *current* vol - trailing 15-30 min realised vol from the 1-second grid, or a GARCH(1,1) on 1-minute bars with variance targeting if that is all you have. Ignore mean reversion at this horizon. Correct for time-of-day.
2. **Shape.** Do not use a Gaussian for the conditional distribution. Student-t with $\nu \approx 4-5$ on the standardised residual reproduces the observed tails; a Gaussian underprices a 4σ state by two orders of magnitude.
3. **Where the edge lives.** Getting the level right is most of the value. The shape matters when the price is already several $\hat{\sigma}$ from the strike with time left: a Gaussian says “99.9%”, the data says “99%”. Whether that residual 1% survives fees is a separate question.

References

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- Andersen, T. & Bollerslev, T. (1997). Intraday periodicity and volatility persistence in financial markets. *J. Empirical Finance*.
- Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *J. Financial Econometrics*.
- Hansen, P., Huang, Z. & Shek, H. (2012). Realized GARCH. *J. Applied Econometrics*.

Code: `analysis/garch/make_figures.py` (*figures, fit, fitted.json*), `analysis/cond_kurt.py` (*realised-vol standardisation*), `analysis/return_dists.py` (*return distributions*).